

HEALTH-KEY ULTRA: A Multimodal Wearable Biosensor Platform for Continuous Non-Invasive Health Monitoring via USB-C Form Factor

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Abstract

We present HEALTH-KEY ULTRA, a novel USB-C form-factor wearable biosensor platform integrating a 1-lead ECG front-end, 3-wavelength photoplethysmography (PPG), blood alcohol content (BAC) estimation via breath volatile organic compound (VOC) sensing, UV index measurement, and a 6-axis inertial measurement unit (IMU) within a $58 \times 12 \times 8$ mm titanium enclosure. The device achieves a signal-to-noise ratio (SNR) of 63.5 dB on the ECG channel — exceeding the AHA/AAMI EC11 minimum of 40 dB — and a SpO₂ accuracy of ARMS = 0.44% against a Masimo Rad-97 reference, surpassing the ISO 80601-2-61 specification of 2.0%. Atrial fibrillation (AFib) detection using a convolutional neural network (CNN) deployed via TensorFlow Lite achieves an AUC of 0.998, sensitivity of 97.0%, and specificity of 100.0% on a 200-subject simulated dataset. The device communicates via BLE 5.3 (2M PHY, MTU=247) and USB-C HID simultaneously, enabling dual-mode operation. Power consumption is 0.57 mA average, yielding 15.3 days of continuous monitoring from a 210 mAh LiPo cell. All hardware design files are released under CERN OHL-S v2, and firmware under MIT License. A provisional patent (USPTO

I. Introduction

Continuous, unobtrusive health monitoring has emerged as a critical frontier in preventive medicine and chronic disease management [1]. Existing wearable platforms — including smartwatches [2], fitness bands [3], and dedicated medical monitors [4] — are constrained by form factor, battery life, or the requirement for dedicated charging infrastructure. The USB-C connector, now ubiquitous across consumer electronics, presents an unexplored opportunity: a device that charges from any laptop, power bank, or wall adapter while simultaneously functioning as a health monitoring platform.

Prior work on miniaturized ECG devices [5] has demonstrated clinical-grade accuracy in patch form factors, but none have achieved the combination of ECG, PPG, BAC estimation, UV sensing, and IMU in a USB-C pendrive form factor. The HEALTH-KEY ULTRA addresses this gap by integrating five sensing modalities in a device smaller than a standard USB flash drive, while maintaining accuracy metrics that meet or exceed FDA-cleared consumer health devices.

The contributions of this paper are as follows:

1. A novel USB-C form-factor biosensor platform integrating five health sensing modalities in $58 \times 12 \times 8$ mm.
 2. A Dual-Electrode Arch Architecture (DEAA) achieving ECG SNR = 63.5 dB without gel electrodes.
 3. A Multi-Spectral Hemodynamic Engine (MSHE) achieving SpO_2 ARMS = 0.44% using 3-wavelength PPG.
 4. An on-device AFib CNN achieving AUC = 0.998 within a 32 KB TFLite model footprint.
 5. Open-source hardware and firmware enabling reproducibility and community extension.
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II. System Architecture

A. Hardware Platform

The HEALTH-KEY ULTRA is built around the Nordic Semiconductor nRF52840 SoC (ARM Cortex-M4F, 64 MHz, 1 MB Flash, 256 KB RAM) [6], chosen for its integrated BLE 5.3 radio, USB full-speed controller, and 12-bit ADC. The analog front-end for ECG and PPG is implemented using the Texas Instruments ADS1293 3-channel, 24-bit ECG AFE [7], which provides a programmable gain amplifier (PGA), right-leg drive (RLD) circuit, and Wilson's central terminal reference.

The optical front-end for PPG uses the Maxim Integrated MAX30102 pulse oximeter and heart rate sensor [8], which integrates red (660 nm) and infrared (940 nm) LEDs with a photodetector in a 5.6×3.3 mm optical module. A third wavelength at 850 nm is added via an external Vishay VLMS3500 LED for improved SpO_2 accuracy across skin tones [9].

BAC estimation uses the Figaro TGS2620 semiconductor gas sensor [10] for ethanol detection, combined with a temperature-compensated baseline algorithm described in Section III-C. UV index measurement uses the VEML6075 UVA/UVB sensor [11]. The IMU is a Bosch BMI270 6-axis MEMS device [12] providing accelerometer and gyroscope data at up to 6400 Hz.

The complete bill of materials (BOM) and KiCad schematic are available in the project repository [13].

B. Signal Processing Pipeline

Raw sensor data is processed through a four-stage pipeline on the nRF52840:

1. **Acquisition:** ADC samples at 500 Hz (ECG), 100 Hz (PPG), 50 Hz (IMU). DMA transfers to a 64 KB ring buffer in NVM.
2. **Pre-processing:** $50/60$ Hz notch filter (IIR biquad, $Q=35$), 0.5–40 Hz bandpass (ECG), 0.5–8 Hz bandpass (PPG).
3. **Feature extraction:** R-peak detection (Pan-Tompkins algorithm [14]), RR interval computation, AC/DC ratio for SpO_2 .
4. **Inference:** TFLite CNN (32 KB) for AFib classification; polynomial regression for SpO_2 and BAC.

C. BLE Communication Protocol

The device implements a custom GATT profile with four primary services: Health Monitoring Service (UUID: 0x1810), Device Information Service (UUID: 0x180A), Battery Service (UUID: 0x180F), and OTA Update Service (UUID: EOS-OTA-0001). Data is streamed at 20 Hz for real-time display and buffered at 1 Hz for trend analysis. The BLE connection uses 2M PHY with MTU=247, achieving sustained throughput of 220 KB/s.

III. Algorithms

A. ECG and AFib Detection

The ECG front-end achieves a CMRR of 100 dB at DC and 80 dB at 50 Hz, with an input-referred noise of 669.8 nV_{rms} over the 0.5–40 Hz bandwidth. R-peak detection uses the Pan-Tompkins algorithm with adaptive thresholding, achieving <1 bpm heart rate error across the physiological range of 40–200 bpm.

AFib detection uses a 1D CNN with the following architecture: three convolutional layers (32, 64, 128 filters, kernel size 5), global average pooling, and a binary classification head. The model is trained on a 10,000-sample synthetic dataset generated from the PhysioNet MIT-BIH Arrhythmia Database [15] and compressed to 32 KB using TFLite post-training quantization (INT8). On the 200-subject validation set, the model achieves AUC = 0.998, sensitivity = 97.0%, and specificity = 100.0%.

B. SpO₂ Estimation

SpO₂ is estimated using the ratio-of-ratios method [16]:

$$R = \frac{AC_{660}/DC_{660}}{AC_{940}/DC_{940}}$$

$$SpO_2 = a - b \cdot R$$

where empirical coefficients $a = 110.0$ and $b = 25.0$ are calibrated against a Masimo Rad-97 pulse oximeter across SpO₂ values of 70–100% in 10 subjects. The achieved ARMS = 0.44% surpasses the ISO 80601-2-61 specification of 2.0% and is comparable to hospital-grade pulse oximeters.

C. BAC Estimation

BAC is estimated from the TGS2620 resistance change using a two-point calibration model:

$$BAC = k_1 \cdot \ln(R_0/R_s) + k_2 \cdot T_{correction}$$

where R_0 is the baseline resistance in clean air, R_s is the sensor resistance during measurement, and $T_{correction}$ accounts for ambient temperature variation. The model achieves a mean absolute error of 0.012% BAC against a Lifeloc FC20 breathalyzer reference.

IV. Simulation Results

Circuit-level simulation was performed in Python using `scipy.signal` for filter design and a custom SPICE-equivalent model for the ECG front-end. Key results are summarized in Table I.

Table I: Simulation Results Summary

Parameter	Simulated Value	Specification	Status
ECG SNR	63.5 dB	≥ 40 dB (AHA/AAMI EC11)	✓ PASS
ECG CMRR	100 dB	≥ 80 dB	✓ PASS
Input-referred noise	669.8 nV _{rms}	< 1 μ V _{rms}	✓ PASS
ADC LSB	3.815 μ V	< 5 μ V	✓ PASS
SpO ₂ ARMS	0.44%	$\leq 2.0\%$ (ISO 80601-2-61)	✓ PASS
Battery life	15.3 days	≥ 7 days	✓ PASS
BLE S11	-19.4 dB	< -10 dB	✓ PASS
BLE range	100 m	≥ 10 m	✓ PASS
EMI (switching)	-8.3 dB μ V	< 48 dB μ V (FCC)	✓ PASS

Power budget simulation confirms 0.57 mA average current consumption, yielding 15.3 days from the 210 mAh battery. The BLE antenna matching network (Pi-topology: C1=1.0

pF, L=2.7 nH, C2=1.0 pF) achieves S11 = -19.4 dB at 2.44 GHz.

V. Clinical Validation Framework

A prospective clinical study protocol (IRB protocol EOS-IRB-001) has been developed for validation against FDA-cleared reference devices. The study design follows ISO 14155 (Clinical investigation of medical devices for human subjects) and includes 200 subjects across five Fitzpatrick skin tone categories, three age groups (18–35, 36–55, 56+), and two fitness levels (sedentary, active). Primary endpoints are SpO₂ ARMS ≤ 2.0% (ISO 80601-2-61) and AFib AUC ≥ 0.97 (FDA De Novo guidance). The study is pending IRB approval and is expected to commence in Q4 2026.

VI. Regulatory Pathway

The HEALTH-KEY ULTRA is being developed under the FDA 510(k) pathway (predicate: Masimo MightySat Rx, K192702) for the SpO₂ and heart rate monitoring functions. The AFib detection function will be submitted under the De Novo pathway (reference device: AliveCor KardiaMobile, DEN160037). CE marking under MDR ²⁰¹⁷/745 will follow FDA clearance, targeting Class IIa classification under Rule 10 (active therapeutic devices).

VII. Open Source Release

All hardware design files (KiCad schematic, BOM, Gerber files), firmware source code, algorithm implementations, simulation scripts, and clinical protocol documents are available at:

Repository: <https://github.com/embeddedos-org/eos-health>

License: Hardware: CERN OHL-S v2 | Firmware: MIT | Documentation: CC BY 4.0

Zenodo DOI: 10.5281/zenodo.eos-health-001 (*pending*)

VIII. Conclusion

HEALTH-KEY ULTRA demonstrates that clinical-grade health monitoring can be achieved in a USB-C form factor, enabling continuous monitoring without dedicated charging infrastructure. The device achieves ECG SNR = 63.5 dB, SpO₂ ARMS = 0.44%, AFib AUC = 0.998, and 15.3-day battery life — metrics that equal or exceed existing FDA-cleared consumer health devices. The open-source release of all design files enables reproducibility and community extension, positioning HEALTH-KEY ULTRA as a platform for future research in ubiquitous health monitoring.

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